

Referring-Expression Model Evaluation

Gearbox part-identification task · 288 samples, 26 task-graph-style dataset labels · 13 models evaluated

Methodology

Each model receives a system prompt containing the full task description (colors/shapes/rows, spatial conventions, part terminology), the list of 26 allowed dataset labels, and one referring expression at a time. The model must return exactly one label. Two grading tiers are reported:

- **Lenient accuracy** — the raw answer is run through a normalizing parser (case-insensitive match, JSON-unwrapping, single-label extraction from prose) before comparing to the target label. Answers the question “did the model identify the correct part?”
- **Strict accuracy** — the raw answer, trimmed of whitespace only, must exactly equal the target label. Answers the question “did the model follow the output-format instruction to the letter?”

Three of the 26 labels (BEARING, PIN, SCREW_ROW1–SCREW_ROW4) are deliberately grouped: they cover several row/side-specific physical parts that render identically and share one dataset label. The system prompt explicitly instructs models to use the group label rather than a specific row/side identifier for these three.

Leaderboard

Model	Hugging Face ID	Lenient Acc	Strict Acc	Invalid	Mean s/sample	Total time
gemma3-12b	google/gemma-3-12b-it	98.6%	97.2%	0	2.316s	11.1 min
qwen3-vl-8b	Qwen/Qwen3-VL-8B-Instruct	98.6%	94.1%	0	0.936s	4.5 min
qwen25-7b	Qwen/Qwen2.5-7B-Instruct	95.8%	89.2%	0	0.806s	3.9 min
gemma4-e4b	google/gemma-4-E4B-it	93.8%	92.7%	2	1.098s	5.3 min
qwen25-vl-7b	Qwen/Qwen2.5-VL-7B-Instruct	89.9%	79.5%	1	0.805s	3.9 min
qwen3-8b	Qwen/Qwen3-8B	89.2%	0.0%	11	15.544s	74.6 min
qwen3-vl-4b	Qwen/Qwen3-VL-4B-Instruct	89.2%	82.6%	1	0.632s	3.0 min
qwen3-4b	Qwen/Qwen3-4B-Instruct-2507	85.4%	85.4%	2	0.615s	3.0 min

Model	Hugging Face ID	Lenient Acc	Strict Acc	Invalid	Mean s/sample	Total time
llama32-vision-11b	meta-llama/Llama-3.2-11B-Vision-Instruct	83.3%	82.6%	3	0.870s	4.2 min
gemma4-e2b	google/gemma-4-E2B-it	71.9%	70.1%	11	0.646s	3.1 min
gemma3-4b	google/gemma-3-4b-it	62.8%	62.2%	10	0.861s	4.1 min
qwen25-3b	Qwen/Qwen2.5-3B-Instruct	62.2%	46.9%	14	0.399s	1.9 min
qwen25-vl-3b	Qwen/Qwen2.5-VL-3B-Instruct	52.8%	35.8%	86	0.429s	2.1 min

Invalid-Output Analysis

An answer is INVALID when it cannot be normalized to any of the 26 allowed labels at all — distinct from a wrong-but-valid answer. Only models with at least one invalid output are shown below.

gemma4-e4b (2/288 invalid, 0.7%)

2 — Invented a nonexistent "right gear" on Row 1/4 (which only have a left gear) instead of the correct gear STAND

#	Referring expression	Raw model output
#66	"right bracket of the white circle row" target: STAND_ROW1_RIGHT	GEAR_ROW1_RIGHT
#257	"right bracket of the blue square row" target: STAND_ROW4_RIGHT	GEAR_ROW4_RIGHT

qwen25-vl-7b (1/288 invalid, 0.3%)

1 — Invented a nonexistent "right gear" on Row 1/4 (which only have a left gear) instead of the correct gear STAND

#	Referring expression	Raw model output
#257	"right bracket of the blue square row" target: STAND_ROW4_RIGHT	GEAR_ROW4_RIGHT

qwen3-8b (11/288 invalid, 3.8%)

8 — Other / unclassified

#	Referring expression	Raw model output
#23	"the handle immediately to the right of the right white gear stand" target: CRANK_HANDLE_ROW1	<think> Okay, let's tackle this problem. The user is asking for the single physical part referred to by the expression "the handle immediately to the ...
#58	"the stand immediately to the left of the crank handle" target: STAND_ROW1_RIGHT	<think> Okay, let's tackle this problem. The user is asking for the physical part referred to by the expression "the stand immediately to the left of ...
#76	"Row 1 gear" target: GEAR_ROW1_LEFT	<think> Okay, let's tackle this problem. The user is asking about the referring expression "Row 1 gear." I need to figure out which single physical pa...

3 — Invented a nonexistent "right gear" on Row 1/4 (which only have a left gear) instead of the correct gear STAND

#	Referring expression	Raw model output
#267	"Row 4 gear" target: GEAR_ROW4_LEFT	<think> Okay, let's tackle this problem. The user is asking about the referring expression "Row 4 gear." I need to figure out which single physical pa...
#268	"blue gear in the square row" target: GEAR_ROW4_LEFT	<think> Okay, let's tackle this problem. The user's referring expression is "blue gear in the square row." I need to figure out which single physical ...
#269	"square-row blue gear" target: GEAR_ROW4_LEFT	<think> Okay, let's tackle this problem. The user's referring expression is "square-row blue gear." I need to figure out which single physical part th...

qwen3-vl-4b (1/288 invalid, 0.3%)

1 — Invented a nonexistent "right gear" on Row 1/4 (which only have a left gear) instead of the correct gear STAND

#	Referring expression	Raw model output
#256	"blue right gear stand in the square row" target: STAND_ROW4_RIGHT	GEAR_ROW4_RIGHT

qwen3-4b (2/288 invalid, 0.7%)

2 — Invented a nonexistent "right gear" on Row 1/4 (which only have a left gear) instead of the correct gear STAND

#	Referring expression	Raw model output
#62	"white right stand" target: STAND_ROW1_RIGHT	GEAR_ROW1_RIGHT
#65	"white right gear stand in the circle row" target: STAND_ROW1_RIGHT	GEAR_ROW1_RIGHT

llama32-vision-11b (3/288 invalid, 1.0%)

3 — Invented a nonexistent "right gear" on Row 1/4 (which only have a left gear) instead of the correct gear STAND

#	Referring expression	Raw model output
#62	"white right stand" target: STAND_ROW1_RIGHT	GEAR_ROW1_RIGHT
#64	"right gear stand of Row 1" target: STAND_ROW1_RIGHT	GEAR_ROW1_RIGHT
#65	"white right gear stand in the circle row" target: STAND_ROW1_RIGHT	GEAR_ROW1_RIGHT

gemma4-e2b (11/288 invalid, 3.8%)

11 — Invented a nonexistent "right gear" on Row 1/4 (which only have a left gear) instead of the correct gear STAND

#	Referring expression	Raw model output
#57	"the right-hand support in circle-marked Row 1" target: STAND_ROW1_RIGHT	GEAR_ROW1_RIGHT
#62	"white right stand" target: STAND_ROW1_RIGHT	GEAR_ROW1_RIGHT
#63	"Row 1 right bracket" target: STAND_ROW1_RIGHT	GEAR_ROW1_RIGHT

gemma3-4b (10/288 invalid, 3.5%)

10 — Invented a nonexistent "right gear" on Row 1/4 (which only have a left gear) instead of the correct gear STAND

#	Referring expression	Raw model output
#55	"the Row 1 right gear stand" target: STAND_ROW1_RIGHT	GEAR_ROW1_RIGHT
#57	"the right-hand support in circle-marked Row 1" target: STAND_ROW1_RIGHT	GEAR_ROW1_RIGHT
#64	"right gear stand of Row 1" target: STAND_ROW1_RIGHT	GEAR_ROW1_RIGHT

qwen25-3b (14/288 invalid, 4.9%)

7 — Invented a nonexistent "right gear" on Row 1/4 (which only have a left gear) instead of the correct gear STAND

#	Referring expression	Raw model output
#55	"the Row 1 right gear stand" target: STAND_ROW1_RIGHT	GEAR_ROW1_RIGHT
#57	"the right-hand support in circle-marked Row 1" target: STAND_ROW1_RIGHT	GEAR_ROW1_RIGHT
#64	"right gear stand of Row 1" target: STAND_ROW1_RIGHT	GEAR_ROW1_RIGHT

6 — Used "GEAR_STAND_ROW{n}_{side}" instead of the canonical "STAND_ROW{n}_{side}"

#	Referring expression	Raw model output
#62	"white right stand" target: STAND_ROW1_RIGHT	GEAR_STAND_ROW2_RIGHT
#119	"red right stand" target: STAND_ROW2_RIGHT	GEAR_STAND_ROW2_RIGHT
#174	"green left stand" target: STAND_ROW3_LEFT	GEAR_STAND_ROW3_LEFT

1 — Model output was empty or a literal non-answer (e.g. "INVALID_LABEL")

#	Referring expression	Raw model output
#22	"the white bent lever with a rounded grip" target: CRANK_HANDLE_ROW1	INVALID_LABEL

qwen25-vl-3b (86/288 invalid, 29.9%)

85 — Answered with a produced-subassembly name (e.g. GEAR_ROD_ROW1_ASSEMBLY) instead of the raw part name

#	Referring expression	Raw model output
#20	“gear-stand bearing” target: BEARING	STAND_ROW1_RIGHT_ASSEMBLY
#23	“the handle immediately to the right of the right white gear stand” target: CRANK_HANDLE_ROW1	CRANK_HANDLE_ROW1_ASSEMBLY
#24	“the crank attached at the right end of the white gear rod” target: CRANK_HANDLE_ROW1	CRANK_HANDLE_ROW1_ASSEMBLY

1 — Invented a nonexistent "right gear" on Row 1/4 (which only have a left gear) instead of the correct gear STAND

#	Referring expression	Raw model output
#257	“right bracket of the blue square row” target: STAND_ROW4_RIGHT	GEAR_ROW4_RIGHT

Strict-vs-Lenient Gap Analysis

These are answers that correctly identify the right part (lenient-correct) but fail literal format compliance (strict-wrong). Only models with at least one such case are shown below.

Key finding: excluding qwen3 - 8b (whose gap is a structural artifact of always emitting a reasoning chain before answering), **100% of every remaining model's strict-accuracy gap** — including the two best-performing models in the study — comes from one specific cause: answering a full row/side-specific identifier (e.g. BEARING_ROW2_LEFT) instead of the grouped label (BEARING) the prompt explicitly requests. This holds regardless of model strength, suggesting the abstraction is genuinely difficult for models to internalize reliably, not a competence gap tied to any one model family or size.

gemma3-12b (4 lenient-correct but strict-wrong, gap = 1.4%)

4 — Answered a specific row/side identifier (e.g. BEARING_ROW2_LEFT) instead of the grouped label (BEARING) the prompt explicitly requests

#	Referring expression	Raw model output
#148	“a screw at the base of a Row 2 gear stand” target: SCREW_ROW2	SCREW_ROW2_LEFT
#272	“a screw at the base of a Row 4 gear stand” target: SCREW_ROW4	SCREW_ROW4_LEFT
#274	“the silver Phillips screw for the blue Row 4 stand” target: SCREW_ROW4	SCREW_ROW4_RIGHT

qwen3-vl-8b (13 lenient-correct but strict-wrong, gap = 4.5%)

13 — Answered a specific row/side identifier (e.g. BEARING_ROW2_LEFT) instead of the grouped label (BEARING) the prompt explicitly requests

#	Referring expression	Raw model output
#81	“a screw at the base of a Row 1 gear stand” target: SCREW_ROW1	SCREW_ROW1_RIGHT
#83	“the silver H5 hex-socket screw for the white Row 1 stand” target: SCREW_ROW1	SCREW_ROW1_RIGHT
#84	“the Row 1 fastener tightened with a 5 mm hex key” target: SCREW_ROW1	SCREW_ROW1_RIGHT

qwen25-7b (19 lenient-correct but strict-wrong, gap = 6.6%)

19 — Answered a specific row/side identifier (e.g. BEARING_ROW2_LEFT) instead of the grouped label (BEARING) the prompt explicitly requests

#	Referring expression	Raw model output
#16	“the black bearing that supports a rotating gear shaft inside a stand” target: BEARING	BEARING_ROW2_LEFT
#18	“black bearing” target: BEARING	BEARING_ROW2_LEFT
#79	“a Row 1 H5 hex-socket mounting screw” target: SCREW_ROW1	SCREW_ROW1_LEFT

gemma4-e4b (3 lenient-correct but strict-wrong, gap = 1.0%)

3 — Answered a specific row/side identifier (e.g. BEARING_ROW2_LEFT) instead of the grouped label (BEARING) the prompt explicitly requests

#	Referring expression	Raw model output
#274	“the silver Phillips screw for the blue Row 4 stand” target: SCREW_ROW4	SCREW_ROW4_RIGHT
#275	“the Row 4 fastener tightened with a Phillips screwdriver” target: SCREW_ROW4	SCREW_ROW4_RIGHT
#276	“the small silver screw located at the foot of a blue support bracket” target: SCREW_ROW4	SCREW_ROW4_LEFT

qwen25-vl-7b (30 lenient-correct but strict-wrong, gap = 10.4%)

30 — Answered a specific row/side identifier (e.g. BEARING_ROW2_LEFT) instead of the grouped label (BEARING) the prompt explicitly requests

#	Referring expression	Raw model output
#18	“black bearing” target: BEARING	BEARING_ROW2_LEFT
#79	“a Row 1 H5 hex-socket mounting screw” target: SCREW_ROW1	SCREW_ROW1_LEFT
#80	“a small silver screw with a recessed H5 hex-socket head” target: SCREW_ROW1	SCREW_ROW1_LEFT

qwen3-8b (257 lenient-correct but strict-wrong, gap = 89.2%)

250 — Model correctly answers after a <think>...</think> block, but strict grading counts the full raw output, which can never equal the bare label

#	Referring expression	Raw model output
#1	"the gearbox mounting plate" target: BASE_BOARD	BASE_BOARD
#2	"the large flat black rectangular plate" target: BASE_BOARD	BASE_BOARD
#3	"the base plate underneath all four colored rows" target: BASE_BOARD	BASE_BOARD

6 — Answered a specific row/side identifier (e.g. BEARING_ROW2_LEFT) instead of the grouped label (BEARING) the prompt explicitly requests

#	Referring expression	Raw model output
#16	"the black bearing that supports a rotating gear shaft inside a stand" target: BEARING	<think> Okay, let's tackle this problem. The user is asking for the single physical part referred to by the expression "the black bearing that support...
#18	"black bearing" target: BEARING	<think> Okay, let's tackle this problem. The user's referring expression is "black bearing." I need to figure out which single physical part this refe...
#276	"the small silver screw located at the foot of a blue support bracket" target: SCREW_ROW4	<think> Okay, let's tackle this problem. The user is asking for the single physical part referred to by the expression: "the small silver screw locate...

1 — Correct label mentioned once inside a longer explanatory sentence

#	Referring expression	Raw model output
#259	"the large blue spur wheel" target: GEAR_ROW4_LEFT	<think> Okay, let's tackle this problem. The user is asking about the referring expression "the large blue spur wheel." I need to figure out which sin...

qwen3-vl-4b (19 lenient-correct but strict-wrong, gap = 6.6%)

19 — Answered a specific row/side identifier (e.g. BEARING_ROW2_LEFT) instead of the grouped label (BEARING) the prompt explicitly requests

#	Referring expression	Raw model output
#81	“a screw at the base of a Row 1 gear stand” target: SCREW_ROW1	SCREW_ROW1_RIGHT
#82	“a fastener used to secure a Row 1 bearing stand to the baseboard” target: SCREW_ROW1	SCREW_ROW1_RIGHT
#83	“the silver H5 hex-socket screw for the white Row 1 stand” target: SCREW_ROW1	SCREW_ROW1_RIGHT

llama32-vision-11b (2 lenient-correct but strict-wrong, gap = 0.7%)

2 — Answered a specific row/side identifier (e.g. BEARING_ROW2_LEFT) instead of the grouped label (BEARING) the prompt explicitly requests

#	Referring expression	Raw model output
#82	“a fastener used to secure a Row 1 bearing stand to the baseboard” target: SCREW_ROW1	SCREW_ROW1_RIGHT
#272	“a screw at the base of a Row 4 gear stand” target: SCREW_ROW4	SCREW_ROW4_LEFT

gemma4-e2b (5 lenient-correct but strict-wrong, gap = 1.7%)

3 — Correct label mentioned once inside a longer explanatory sentence

#	Referring expression	Raw model output
#7	"the broad black rectangular base beneath the white, red, green, and blue mechanisms" target: BASE_BOARD	```text BASE_BOARD ```
#126	"the left-side gear on the shaft in the second row from Row 1" target: GEAR_ROW2_LEFT	```text GEAR_ROW2_LEFT ```
#137	"the right-side gear on the shaft in the second row from Row 1" target: GEAR_ROW2_RIGHT	```text GEAR_ROW2_RIGHT ```

2 — Answered a specific row/side identifier (e.g. BEARING_ROW2_LEFT) instead of the grouped label (BEARING) the prompt explicitly requests

#	Referring expression	Raw model output
#81	"a screw at the base of a Row 1 gear stand" target: SCREW_ROW1	SCREW_ROW1_LEFT
#274	"the silver Phillips screw for the blue Row 4 stand" target: SCREW_ROW4	SCREW_ROW4_LEFT

gemma3-4b (2 lenient-correct but strict-wrong, gap = 0.7%)

2 — Answered a specific row/side identifier (e.g. BEARING_ROW2_LEFT) instead of the grouped label (BEARING) the prompt explicitly requests

#	Referring expression	Raw model output
#18	"black bearing" target: BEARING	BEARING_ROW2_LEFT
#285	"the short brown dowel crossing the gear rod beside a gear or the crank" target: PIN	PIN_ROW2_LEFT

qwen25-3b (44 lenient-correct but strict-wrong, gap = 15.3%)

44 — Answered a specific row/side identifier (e.g. BEARING_ROW2_LEFT) instead of the grouped label (BEARING) the prompt explicitly requests

#	Referring expression	Raw model output
#14	“the bearing through which a gear shaft passes” target: BEARING	BEARING_Row2_LEFT
#15	“the small black ring-shaped bearing that fits into each colored stand” target: BEARING	BEARING_RowX_LEFT
#16	“the black bearing that supports a rotating gear shaft inside a stand” target: BEARING	BEARING_ROW2_LEFT

qwen25-vl-3b (49 lenient-correct but strict-wrong, gap = 17.0%)

49 — Answered a specific row/side identifier (e.g. BEARING_ROW2_LEFT) instead of the grouped label (BEARING) the prompt explicitly requests

#	Referring expression	Raw model output
#11	“the radial shaft-support bearing” target: BEARING	BEARING_ROW1_RIGHT_ASSEMBLY
#12	“the small black circular ring with a center hole” target: BEARING	BEARING_ROW2_LEFT
#13	“the round black part centered inside a bearing support bracket” target: BEARING	BEARING_ROW2_LEFT